Eleanor Wedell

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Education

University of Illinois Urbana-Champaign Urbana-Champaign, IL
Doctor of Philosophy in Computer Science, GPA 3.87 *Aug. 2022 – May 2026 (expected)*

University of Illinois Urbana-Champaign Urbana-Champaign, IL
Master of Science in Computer Science, GPA 3.89 *Aug. 2020 – May 2022*

- Thesis Title: SCAMPP: Scalable Alignment-based Phylogenetic Placement

Drexel University Philadelphia, PA
Bachelor of Science in Mathematics, cum laude, GPA 3.67 *Aug. 2013 – Aug. 2015*

Research summary

I do applied algorithms research in two different areas: network science and computational biology. My MS degree contributed the SCAMPP method for phylogenetic placement (adding sequences into a phylogenetic tree) under maximum likelihood. My PhD work has developed Batch-SCAMPP, an improved version of SCAMPP designed for placing thousands to millions of sequences into the tree and achieving sublinear runtime. I also developed TIPP3, a new method for abundance profiling of reads in microbiome samples, which has superior accuracy to all current methods under most conditions. I work on the problem of community detection, also known as clustering, on large networks, where I have contributed a method for core-periphery community detection (IKC, or Iterative k-core clustering) and an ensemble method (FastEnsemble) for community detection. I am currently working on developing scalable methods for phylogenetic networks construction with theoretical guarantees and a new method for adding sequences into multiple sequence alignments.

Experience

Research Assistant of Professor Tandy Warnow Aug. 2020 – Present
University of Illinois Urbana-Champaign *Urbana-Champaign, IL*

Teaching Assistant Aug. 2020 – Present
University of Illinois Urbana-Champaign *Urbana-Champaign, IL*

- CS 128 (Fall 2023, Spring 2024, Fall 2024) CS 125 (Fall 2020), CS 102 (Spring 2021), and CS 581 (Fall 2021)
- Prepare and conduct weekly labs and quiz/midterm preparatory sessions

C3.ai DTI Curriculum Development Intern April 2021 – May 2021
University of Illinois Urbana-Champaign *Urbana-Champaign, IL*

- Tested the cloud computing platform and provided feedback for instructors at C3.ai Digital Transformation Institute for the development of new AI and machine learning courses introduced by C3.ai

Programmer Analyst Feb. 2016 – Jan. 2020
Innovative Software Solutions, Inc. *Maple Shade, NJ*

- Designed, developed, and maintained custom programs for employer auditing modules for new client installs | Programmed under strict deadlines for accurate reporting for >100K 1099s, W2s, and tax files in year-end team | Enhanced, tested, and debugged code for contributions team, providing annuity distributions and loan processing

Technical Skills

Languages: Python, C/C++, Go, Java, COBOL, Matlab, Bash Scripting
Developer Tools: Linux, Git, VS Code, Neovim
Libraries: OpenMP, MPI, Dendropy, Pytorch, Treeswift, Networkit, igraph, GoTree
Relevant Graduate Coursework: Algorithms, Algorithmic Genomic Biology, Artificial Intelligence, ML for Bioinformatics, Deep Learning, Parallel Programming, Computational Cancer Genomics, Computational Scientometrics

Professional Service and Leadership Roles

Reviewer – Sub-reviewed for RECOMB conference 2023

Supervisor-Undergraduate Research, *UIUC Fall 2023* – Oversee and mentor undergraduate researcher

Journal and Refereed Conference Publications

Wedell E., Park M., Korobskiy D., Warnow T., Chacko G. (2022) Center-Periphery Structure in Communities. *Quantitative Science Studies*, 3(1), 289-314. https://doi.org/10.1162/qss_a_00184.

Wedell E., Cai Y., Warnow T. (2022). SCAMPP: Scaling Alignment-based Phylogenetic Placement to Large Trees. *IEEE/ACM Transactions on Computational Biology and Bioinformatics*, 20(2), 1417-1430. https://doi.org/10.1109/TCBB.2022.3170386.

Wedell E., Cai Y., Warnow T. (2021) Scalable and Accurate Phylogenetic Placement Using pplacer-XR. *In: Martín-Vide C., Vega-Rodríguez M.A., Wheeler T. (eds) Algorithms for Computational Biology* (AlCoB 2021). https://doi.org/10.1007/978-3-030-74432-8_7.

Wedell E., Shen C., Warnow T. (2023) BATCH-SCAMPP: Scaling Phylogenetic Placement Methods to Place Many Sequences (Abstract). *In 23rd International Workshop on Algorithms in Bioinformatics* (WABI 2023). https://doi.org/10.4230/LIPIcs.WABI.2023.3

Tabatabaee Y., **Wedell E.**, Park M., Warnow T. (2024) FastEnsemble: A new scalable ensemble clustering method. *To appear: 13th International Conference on Complex Networks and their Applications* (CNA 2024). https://arxiv.org/abs/2409.02077

Journal Submissions

Wedell E., Shen C., Warnow T. (2024) BSCAMPP: Batch Scaled Phylogenetic Placement on Large Trees. Submitted and Under Review. https://doi.org/10.1101/2022.10.26.513936

Shen C., **Wedell E.**, Pop M., Warnow T. (2024). TIPP3 and TIPP3-fast: Improved Abundance Profiling in Metagenomics. Submitted and Under Review. https://doi.org/10.1101/2024.10.28.620576.

In progress

Willson, J., **Wedell, E.**, Warnow. T. Fast, scalable, and accurate methods for constructing level-1 phylogenetic networks from SNP data. In progress.

Awards and Fellowships

Siebel Scholar, Class of 2022 – awarded annually for academic excellence and demonstrated leadership

SURGE Fellowship – August 2022 - January 2025

Wing Kai Cheng Fellowship – August 2022 - May 2023

Honorable Mention NSF-GRFP – 2022

Debra and Ira Cohen Graduate Fellowship in Computer Science – Starts January 2025

Software

SCAMPP This is software for phylogenetic placement of sequences into large phylogenies based on maximum likelihood, achieving high accuracy and scalability to very large trees (tested up to 200,000 leaves). | *Python, C++, OpenMP, Treeswift* | https://github.com/chry04/PLUSplacer

BSCAMPP This is an algorithmic modification of SCAMPP to enable it to achieve runtime that is sublinear in the number of sequences that are added to a large phylogeny. | *Python, C++, OpenMP, Treeswift* | https://github.com/ewedell/BSCAMPP

IKC This is code for Iterative k-core clustering, designed for use in finding center-periphery clusters. | *Python, Networkit, Networkx* | https://github.com/chackoge/ERNIE_Plus/tree/master/Illinois/clustering/eleanor/code

FastEnsemble This new scalable ensemble method for consensus clustering is the most accurate method on networks with less obvious clusters. | *C++, Python, Networkx, iGraph* | https://github.com/ytabatabaee/fast-ensemble

TIPP3 This abundance profiling and taxon identification method is the most accurate method on PacBio reads and read not known to the reference database. | *Python, Java, BLAST, Dendropy* | https://github.com/c5shen/TIPP3/tree/main